

Data Loading & Saving in NumPy

Save array (.npy file)

```
In [1]: import numpy as np
arr = np.array([1,2,3,4])
np.save("data.npy",arr)
```

Load array

```
In [2]: arr = np.load("data.npy")
```

```
In [3]: print(arr)
```

```
[1 2 3 4]
```

Multiple arrays save (.npz) and load

```
In [4]: a = np.array([1,2])
b = np.array([3,4])
np.savez("Multiple.npz",a=a,b=b)
```

```
In [6]: data = np.load("Multiple.npz")

print(data["a"])
print(data["b"])
```

```
[1 2]
```

```
[3 4]
```

NumPy Indexing and Selection

```
In [1]: import numpy as np
arr = np.array([[1,2,3],[4,5,6],[7,8,9]])
arr
```

```
Out[1]: array([[1, 2, 3],
               [4, 5, 6],
               [7, 8, 9]])
```

Row index → 0,1,2 Column index → 0,1,2

```
In [11]: print(arr[1,2]) # Single element access
```

```
6
```

```
In [15]: print(arr[1]) # arr[1] -> complete row
```

```
[4 5 6]
```

```
In [3]: arr[:,1] #Complete column
```

```
Out[3]: array([2, 5, 8])
```

```
In [17]: arr[0:2, 1:3] # Slicing, arr[row_start:row_end, col_start:col_end]
```

```
Out[17]: array([[2, 3],
               [5, 6]])
```

Logical Selection (Filtering)

```
In [18]: arr = np.array([10,20,30,40])
```

```
arr > 20
```

```
Out[18]: array([False, False,  True,  True])
```

```
In [19]: arr[arr > 20]
```

```
Out[19]: array([30, 40])
```

```
In [20]: arr[arr % 20 == 0]
```

```
Out[20]: array([20, 40])
```

Broadcasting

Different shapes arrays par automatic operations

```
In [21]: arr = np.array([1,2,3])
```

```
arr + 5
```

```
Out[21]: array([6, 7, 8])
```

```
In [22]: # 2D Broadcasting
```

```
arr = np.array([[1,2,3],
               [4,5,6]])
```

```
arr + np.array([10,20,30])
```

```
Out[22]: array([[11, 22, 33],
               [14, 25, 36]])
```

Practice (Important)

- 3×3 array banao
- Second row print karo

- All elements > 5 filter karo
- Array me +10 broadcast karo
- Array ko save aur load karo